GTS123 Introduction to Computers and Programming

Lab 10: Problem 5

Write a program that accepts coordinates of 3 points, i.e. $P_1 = (x_1, y_1, z_1)$, $P_2 = (x_2, y_2, z_2)$, and $P_3 = (x_3, y_3, z_3)$. The program then finds the **longest distance** between any two points from these three points.

The distance function d between two points $P_i = (x_i, y_i, z_i)$ and $P_j = (x_j, y_j, z_j)$ is defined as:

$$d(P_i, P_j) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}$$

You must use the NumPy arrays and functions to solve this problem.

Hint: You need to calculate $d(P_1, P_2)$, $d(P_2, P_3)$, and $d(P_3, P_1)$, then find the maximum among them.

The output should look similar to the following.

```
Input coordinate of P1: 0 0 0
Input coordinate of P2: 1 0 1
Input coordinate of P3: 5 3 0
Output:
The longest distance = 5.83
```

Note: To pass the test, your program must produce the exact output as shown above, including spaces and line breaks.